

MariaDB 安装指导

基于Docker

2025年3月20日

情况说明

- 由于OceanBase需要6GB的内存，对电脑要求较高，部分同学的电脑可能硬件不支持，因此我们提供一个备选方案。
- 对于运行OceanBase数据库内存不足的同学，可以选择使用mariaDB数据库替换OceanBase数据库进行实验，但我们强烈建议大家在条件允许的前提下尽量选择OceanBase数据库进行实验，这对于之后的比赛有很大的帮助。

Docker安装

- 为了便于后续实验，统一使用Docker搭建数据库环境。
 - 参考之前的指导完成安装

安装MairaDB

- 在命令行 (MacOS: 终端Terminal, Windows: cmd) 中输入语句, 下载镜像并创建容器:

```
penghan@penghandeMacBook-Pro-2 ~ % docker run --detach --name mariadb_demo --env MARIADB_USER=penghan --env MARIADB_PASSWORD=penghan_password --env MARIADB_ROOT_PASSWORD=my-root_password mariadb:latest
```

```
Unable to find image 'mariadb:latest' locally
```

```
latest: Pulling from library/mariadb
```

```
6ea603f1df5e: Pull complete
```

```
ccf83fcdbd3f9: Pull complete
```

```
b1db0fbcea43: Pull complete
```

```
840258738964: Pull complete
```

```
e00d176ef650: Pull complete
```

```
3eae2bb1c643: Pull complete
```

```
b6408b7f2598: Pull complete
```

```
30c8ffa0af98: Pull complete
```

```
Digest: sha256:77e8ad8ae65e4bf59da8e0c078a086fce71a5cff21ae3c72e55573f3fde33580
```

```
Status: Downloaded newer image for mariadb:latest
```

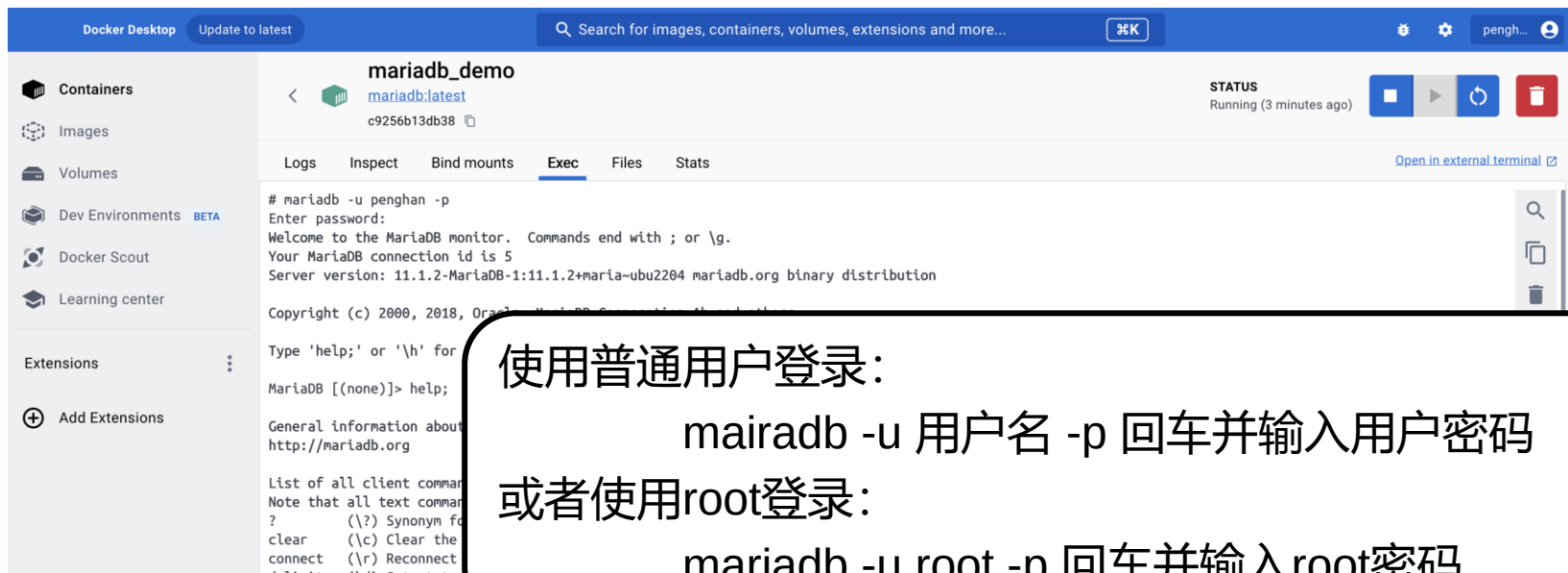
```
c9256b13db386578aba3ceb79e1e9c8037374cd487aad285096d7d7aba84757e
```

```
penghan@penghandeMacBook-Pro-2 ~ %
```

```
docker run --detach --name 容器名称 --env MARIADB_USER=用户名 --env MARIADB_PASSWORD=用户密码 --env MARIADB_ROOT_PASSWORD=root密码 mariadb:latest
```

连接MariaDB并进行实验

- 在Docker中打开该容器，选择Exec后输入下述指令连接MariaDB并开展实验：



The screenshot shows the Docker Desktop interface with the 'mariadb_demo' container selected. The 'Exec' tab is active, showing a terminal window. The terminal output is as follows:

```
# mariadb -u penghan -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 11.1.2-MariaDB-1:11.1.2+maria-ubu2204 mariadb.org binary distribution

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Type 'help;' or '\h' for help; Type '\c' to clear the current input statement.

MariaDB [(none)]> help;

General information about the MariaDB server
http://mariadb.org

List of all client commands
Note that all text commands must be preceded by an exclamation mark !

?      (?)  Synonym for help
clear  (\c) Clear the current input statement
connect (\r) Reconnect to the server
quit   (\q) Quit the MariaDB client
```

Use ordinary user login:

mairadb -u 用户名 -p 回车并输入用户密码

或者使用root登录:

mariadb -u root -p 回车并输入root密码

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